

Final Project Evaluation: How Your Work Is Assessed

The final project is evaluated according to the published project requirements, the course learning objectives, and the quality of the submitted work.

The purpose of the project is to demonstrate a complete and reproducible engineering data analysis workflow. Therefore, the evaluation considers not only whether the code runs, but also whether the analysis is methodologically correct, clearly documented, logically structured, and appropriately interpreted.

Point deductions reflect the extent to which the submitted work fulfils these requirements.

1. Case Study Evaluation – Data Analysis

1.1 Readability and Storytelling of the Analysis Process (3 points)

Your Jupyter Notebook should present the analysis as a clear and understandable process.

A strong submission combines:

- executable code,
- explanatory Markdown text,
- clearly structured sections,
- appropriate visualisations,
- explanations of analytical decisions, and
- interpretations of intermediate and final results.

Code alone is not sufficient. Readers should understand:

- what you are doing,
- why you are doing it,
- what result you expect, and
- what the result means.

Points may be deducted when large parts of the notebook contain code without sufficient explanation or when the overall analytical workflow is difficult to follow.

1.2 Data Import and Data Integration (3 points)

You are expected to import the relevant datasets correctly and integrate them using suitable key variables.

The notebook should explain:

- which datasets are used,
- why they are relevant,
- how they are related,
- which variables are used for merging, and
- how the success of the merge is checked.

Simply importing several files does not fully demonstrate data integration. The relationships between the datasets and the purpose of combining them must also be clear.

Points may be deducted when datasets are imported without a documented selection or integration strategy.

1.3 Data Preparation (3 points)

Before analysing the data, you should systematically examine, clean, and validate it.

Typical data-preparation steps include:

- checking data types,
- identifying missing values,
- detecting duplicates,
- checking inconsistent formats,
- identifying implausible values,
- examining distributions and outliers,
- correcting or documenting data-quality problems, and
- verifying that the cleaning steps were successful.

Exploratory data analysis should normally take place before major transformations or merge operations. This helps you understand the structure and quality of the original data.

Points may be deducted when datasets are directly transformed or merged without first examining and validating them.

1.4 Analytical Approach and Strategy (6 points)

Your analysis should follow a clear and goal-oriented strategy.

A complete analytical approach normally includes:

1. defining the objective of the analysis,
2. introducing the relevant datasets,
3. preparing and validating the data,
4. formulating analytical questions or hypotheses,
5. selecting suitable methods,
6. applying those methods correctly, and
7. interpreting the results in relation to the task.

You should explain why the selected methods are appropriate for the available data and the project objective.

A technically executable notebook does not automatically demonstrate a strong analytical strategy. The purpose and relationship of the individual analytical steps must also be clear.

Points may be deducted when:

- no overall analysis plan is presented,
- the datasets are not introduced,
- individual steps are not connected to the project objective,
- methods are applied without justification, or
- results are not interpreted.

1.5 Understanding of the Assigned Task (4 points)

Your submission should clearly address the objective of the case study.

This includes:

- selecting relevant variables,
- removing or excluding irrelevant information,
- transforming raw data into an analysable dataset,
- completing merge and validation procedures, and
- conducting an analysis that directly contributes to answering the assigned question.

A merged dataset is not automatically analysis-ready. It should be checked for:

- duplicated columns,

- unresolved merge variables,
- unexpected missing values,
- inconsistent records, and
- incomplete data integration.

Points may be deducted when the final dataset remains incomplete, inconsistent, or unsuitable for the intended analysis.

1.6 Best Coding Practices (1 point)

The submitted code should be readable, structured, maintainable, and reproducible.

Good coding practices include:

- meaningful variable and function names,
- logical organisation of code cells,
- reusable functions for repeated operations,
- limited code duplication,
- appropriate comments,
- correct use of libraries, and
- clear separation of the main analytical steps.

Defining a small number of functions is not sufficient when the remaining workflow is highly repetitive or unstructured.

Points may be deducted when the code is difficult to understand, unnecessarily repetitive, or not organised in a way that another analyst could reproduce or maintain.

2. Case Study - Web Application Evaluation

The web application is evaluated according to the quality of its visualisations, functionality, content, design, and the documentation and organisation of the submitted files.

2.1 Visualizations (3 points)

The visualisations should present the analysis results clearly, correctly, and meaningfully.

A strong submission includes:

- suitable visualisation methods for the respective data,

- clear titles, labels, legends, and units,
- informative and understandable charts,
- appropriate interactive elements,
- filters or controls that help users explore the results, and
- interactions that provide additional meaningful information.

The selected chart types should match the data and the analytical question.

Interactivity should support the interpretation of the results rather than being included only for technical demonstration.

Points may be deducted when:

- unsuitable chart types are used,
- charts are unclear or misleading,
- labels, units, or explanations are missing,
- visualisations do not provide meaningful information,
- interactive elements do not work, or
- interactions do not add value to the analysis.

2.2 Functionality (4 points)

The submitted web application should run correctly and provide meaningful, working functions.

A functional application should:

- start without errors,
- load the required data correctly,
- display all relevant components,
- respond correctly to user input,
- contain working buttons, filters, menus, and other controls, and
- produce appropriate outputs based on the selected interactions.

The number of functions alone does not determine the quality of the application. The functions should be relevant to the task and should operate reliably.

Points may be deducted when:

- the application does not start,

- errors occur during normal use,
- important functions are incomplete,
- buttons or filters do not work,
- the application produces incorrect or inconsistent results, or
- the available functions do not contribute meaningfully to the project objective.

2.3 Content (4 points)

The information presented in the web application should be directly related to the assigned task and the results of the data analysis.

A strong application presents:

- relevant analytical results,
- appropriate key figures,
- meaningful comparisons,
- useful explanatory information, and
- interactions that are logically connected to the displayed content.

The application should help users understand the most important findings of the project. The content should not consist only of disconnected charts or technical elements.

Points may be deducted when:

- the displayed information is unrelated to the task,
- important results are missing,
- information is presented without context,
- interactions do not correspond to the displayed data, or
- the application does not communicate a clear analytical message.

2.4 Design (1 point)

The application should have a clear, consistent, and user-friendly appearance.

The design evaluation considers:

- layout and structure,
- readability,
- visual consistency,

- navigation,
- spacing and alignment,
- appropriate use of colours,
- clarity of controls, and
- the overall level of effort demonstrated in the application.

A technically working application may still lose points when its structure or appearance makes it difficult to use or understand.

Points may be deducted when:

- the layout is confusing,
- elements overlap or are poorly aligned,
- text or visualisations are difficult to read,
- colours are inconsistent or unsuitable,
- navigation is unclear, or
- the application appears incomplete or insufficiently developed.

2.5 Submission, File Organisation, and Documentation (3 points)

All submitted files should be complete, clearly named, properly organised, and sufficiently documented.

The submission should include:

- all required notebooks,
- HTML exports of the notebooks,
- the web application file,
- the final processed dataset when required,
- any additional files required to run the project, and
- clear instructions when additional setup is necessary.

The submitted files should follow the required naming conventions and folder structure. The notebooks and application should use relative file paths so that the project can be run on another computer.

The original raw datasets should generally not be uploaded when they have already been provided for the project. Uploading unnecessary original data increases the submission size and may result in a point deduction.

Points may be deducted when:

- required files are missing,
- the final processed dataset is missing,
- files required to run the application are not included,
- unnecessary original datasets are uploaded,
- file names do not follow the required format,
- the folder structure is unclear,
- the submission is unnecessarily large,
- absolute or computer-specific file paths are used,
- notebooks are not run before submission,
- outputs are missing from the submitted notebooks or HTML files, or
- the submitted files do not contain enough documentation to understand and reproduce the project.

The evaluation considers whether another person can identify the relevant files, understand their purpose, and run the complete project without having to reconstruct the submission structure.